Pyramid Surface Area Build

Unit 10: Time Capsule Engineers · Lesson 10-5 · EduWonderLab

UNIT 10 · LESSON 10-5 · TEACHER NOTES | **Standard 6.G.4**

FORMAT **Build a net + calculate**	TIME **25–35 min**
GROUPING **Pairs**	TOPIC **Surface Area of Pyramids**
STANDARD **6.G.4**	PREP LEVEL **Medium — print pyramid net templates**

Learning goal *I can find the surface area of a square pyramid by adding the base and triangular faces.*

Teacher setup

- Print one template set per pair. Students build a pyramid net (square base + 4 triangles).
- They find the base area and the four triangular faces, then combine.
- Each triangle uses ½ × base × slant height.

Materials

Pyramid net templates, Ruler, Base/lateral area organizer, Pencil.

Differentiation & language support

- Organizer separates base area from lateral (triangle) area (SPED).
- Sentence stem: "Surface area includes ___ because ___."
- ESOL: label the slant height on the triangle.

Key vocabulary (English / Español):

Term	Español	What it means
pyramid	pirámide	a solid with a base and triangular faces
slant height	altura inclinada	the height of a triangular face

Quick teacher check: *Check one triangular-face area, e.g., Card 2 (½ × 6 × 5 = 15).*

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Name: **Date:** **Class:**

Your goal: *I can find the surface area of a square pyramid by adding the base and triangular faces.*

What you need

Pyramid net templates, Ruler, Area organizer.

How to play

1. Pick a pyramid card and find the base area.
2. Find the area of one triangular face (½ × base × slant height).
3. Multiply by 4 for all four triangles.
4. Add the base and triangles for the surface area.

Pyramid Cards

Add the square base to the four triangular faces. Each triangle = ½ × base × slant height.

1. Square base 4 × 4, slant height 6	**2.** Square base 6 × 6, slant height 5
3. Square base 3 × 3, slant height 4	**4.** Square base 5 × 5, slant height 8
5. Square base 2 × 2, slant height 3	**6.** A pyramid has base area 16 and 4 triangles totaling 48. Surface area?

Area organizer (base + 4 triangles):

Explain your thinking

Explain why surface area includes the base and all four triangular faces.

Sentence frame: *Surface area includes ___ because ___.*

★ **Challenge:** *Card 1 has surface area 64. If the slant height doubled to 12, what is the new surface area? (16 + 96 = 112)*

ANSWER KEY

Teacher copy — please do not distribute to students.

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Answers

1. Square base 4 × 4, slant height 6 **Answer: 64 sq units (16 + 48)**	**2.** Square base 6 × 6, slant height 5 **Answer: 96 sq units (36 + 60)**
3. Square base 3 × 3, slant height 4 **Answer: 33 sq units (9 + 24)**	**4.** Square base 5 × 5, slant height 8 **Answer: 105 sq units (25 + 80)**
5. Square base 2 × 2, slant height 3 **Answer: 16 sq units (4 + 12)**	**6.** A pyramid has base area 16 and 4 triangles totaling 48. Surface area? **Answer: 64 sq units**

Teaching notes & look-fors

- Surface area = base area + 4 × (½ × base × slant height).
- Card 1: 16 + 4(½ × 4 × 6) = 16 + 48 = 64.